

Session 11

1. Pick any one feature from an app you use daily (for example: 'Search Restaurants' in Zomato or 'Add to Cart' in Flipkart) and break down the testing tasks for this feature using a Work Breakdown Structure (WBS) — list at least 6 distinct testing activities or sub-tasks.

Work Breakdown Structure (WBS) – “Search Restaurants” in Zomato

Feature selected: Search Restaurants in Zomato

A Work Breakdown Structure (WBS) divides a testing task into smaller activities.

No.	Testing Activity / Sub-task	Description
1	Requirement Analysis	Understand how restaurant search should work.
2	Test Scenario Identification	Identify different search scenarios such as restaurant name, food name, and location.
3	Test Case Design	Create test cases with steps, test data, and expected results.
4	Functional Testing	Check whether the search returns correct restaurants and food items.
5	Negative Testing	Test invalid, blank, misspelled, and unavailable searches.
6	UI Testing	Check search box, buttons, filters, text, and layout.
7	Compatibility Testing	Test the search feature on different browsers and mobile devices.
8	Defect Reporting	Report any defects found during testing.
9	Retesting	Test the fixed defects again.
10	Regression Testing	Check that changes have not affected other features.

WBS structure:

Search Restaurants Testing
|
|—— Requirement Analysis

- Test Scenario Identification
 - Test Case Design
 - Functional Testing
 - Negative Testing
 - UI Testing
 - Compatibility Testing
 - Defect Reporting
 - Retesting
 - Regression Testing
-

2. For the 'Send Message' feature in WhatsApp, estimate the total test effort required by listing all the main test activities, assigning estimated hours to each, and calculating the overall effort needed.

 Hint: Consider activities like test case design, test execution, defect reporting, and regression testing.

Test Effort Estimation – WhatsApp “Send Message”

Feature: Send Message

For this estimation, we assume that one tester is testing the normal text-message functionality.

No.	Test Activity	Estimated Hours
1	Requirement Analysis	2 hrs
2	Test Scenario Identification	2 hrs
3	Test Case Design	5 hrs
4	Test Data Preparation	1 hr
5	Test Environment Setup	1 hr
6	Test Execution	6 hrs
7	Defect Reporting	2 hrs
8	Retesting Fixed Defects	2 hrs

9	Regression Testing	3 hrs
10	Test Summary Report	1 hr
Total Estimated Effort		25 hrs

Assumptions

- Testing is performed by **one tester**.
- The feature is limited to **sending text messages**.
- Basic positive and negative scenarios are included.
- Testing is performed on commonly used mobile devices.
- Development time is not included.
- Major performance/security testing is not included.

Calculation

Total Effort = 2 + 2 + 5 + 1 + 1 + 6 + 2 + 2 + 3 + 1

Total Estimated Testing Effort = 25 hours

3. Given the following scenario: Flipkart's 'Apply Coupon' feature is being released as a small module. Using the steps of test effort estimation discussed in class, create a simple estimation sheet (table) showing each test activity, estimated hours, and any assumptions you make.

Test Effort Estimation – Flipkart “Apply Coupon”

Feature: Apply Coupon

The following is a simple estimation sheet for testing the Apply Coupon module.

No	Test Activity	Estimated Hours	Assumptions
1	Requirement Analysis	1 hr	Coupon rules and conditions are available.

2	Test Scenario Identification	2 hrs	Valid, invalid, expired and minimum-order cases are considered.
3	Test Case Design	4 hrs	Around 20–25 test cases are created.
4	Test Data Preparation	1 hr	Different coupon codes and cart values are prepared.
5	Test Environment Setup	1 hr	Application and test account are available.
6	Functional Test Execution	5 hrs	All prepared test cases are executed.
7	Negative Testing	2 hrs	Invalid, expired and already-used coupons are tested.
8	Defect Reporting	1 hr	Defects are documented with evidence.
9	Retesting	2 hrs	Fixed defects are tested again.
10	Regression Testing	2 hrs	Cart and payment-related functionality is checked.
11	Test Summary Report	1 hr	Testing results are documented.
Total		22 hrs	

Total Estimated Effort

22 hours

Important assumptions

1. The Apply Coupon module is small and has clearly defined requirements.
2. Test environment and test accounts are already available.
3. Coupon rules are provided by the business team.
4. No major changes are expected during testing.
5. Performance and security testing are outside this estimation.

4. Read about Test Point Analysis (TPA) as introduced in class. In your own words, explain how TPA could help estimate testing effort for a new 'Movie Booking' feature in BookMyShow. List two advantages and one limitation of using

TPA for this purpose.

Hint:
You do not need to do full TPA calculations, just explain the
concept and its application.

Test Point Analysis (TPA) – BookMyShow “Movie Booking”

What is Test Point Analysis?

Test Point Analysis (TPA) is a technique used to estimate the amount of testing effort required for a software feature.

It considers factors such as:

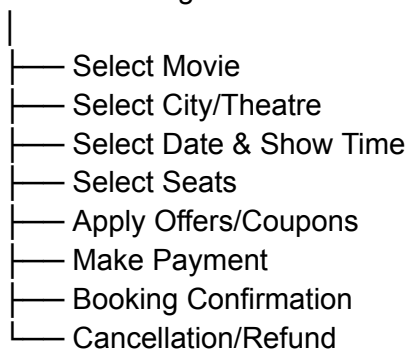
- Size and complexity of the functionality
- Number of test conditions
- Different interfaces and inputs
- Complexity of the system
- Testing requirements

The estimated test points can then be converted into an approximate **testing effort**, such as tester-hours.

Applying TPA to Movie Booking

For a new **Movie Booking** feature in BookMyShow, we can identify different functional areas such as:

Movie Booking



Each area can have different complexity. For example, **seat selection and payment** may require more testing effort because they involve more conditions and possible failure scenarios.

TPA can be used to estimate the relative size of these testing requirements and convert them into an approximate number of tester-hours.

Two Advantages of TPA

1. Structured estimation

TPA provides a systematic way to estimate testing effort instead of simply guessing the number of hours.

2. Helps with resource planning

The estimated effort can help a Test Leader decide how many testers and how much time may be required.

One Limitation of TPA

TPA depends on the quality of the information available.

If the requirements or complexity estimates are inaccurate, the final testing effort estimate may also be inaccurate.

Final Assignment Summary

Question	Topic	Answer
1	WBS – Search Restaurants	10 testing activities
2	WhatsApp Send Message	25 hours estimated effort
3	Flipkart Apply Coupon	22 hours estimated effort
4	Test Point Analysis	Concept + application + 2 advantages + 1 limitation